

24/24

Excellent!

PHY201 Final Exam on chapter(s): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

Name: \_\_\_\_\_

100%

These questions assess your understanding of the material from Chapter(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Suppose that there is heat transfer of 46 J to a system, while the system does 51 J of work. Later, there is heat transfer of 42 J out of the system while 49 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: 2.0 J ✓

$$\Delta U = Q - W$$

$$\Delta U = (46\text{ J} - 42\text{ J}) - (51\text{ J} - 49\text{ J}) = 2.0\text{ J}$$

2. Tyler is not only a part-time guide at the Battleship USS Iowa Museum, but also a physics student. He wants to find the net external force that would be exerted on a 1170-kg artillery shell fired from one of the USS Iowa's turrets. Tyler measures the barrel of a turret to be 16.2-m long and reads that a shell, fired from rest, would leave the barrel at 670 m/s. What is the net external force exerted on the shell?

ANSWER:  $1.621 \times 10^7 \text{ N}$  ✓

$$\vec{F} = m\vec{a}$$

$$v^2 = v_0^2 + 2\vec{a}\Delta\vec{x}$$

$$\vec{a} = \frac{v^2}{2\Delta x} \quad \because v_0 = 0$$

$$\vec{F} = \frac{mv^2}{2\Delta x} = \frac{1170\text{ kg} \cdot 670^2 \frac{\text{m}^2}{\text{s}^2}}{2 \cdot 16.2\text{ m}} = 1.621 \times 10^7 \text{ N}$$

3. Grace has to push a 280-kg crate made of ironbark maple on a floor made of verge willow over a distance of 11 m. The crate is initially at rest on the horizontal floor, until Grace pushes horizontally on the crate with the minimum force necessary to start its motion. If Grace continues pushing on the crate with this force after the crate begins to move, what is the magnitude of the crate's resulting acceleration? Note that for ironbark maple on verge willow,  $\mu_s = 0.05$  and  $\mu_k = 0.05$ .

$$F_g = F_{g\text{roce}}$$

ANSWER:  $0 \frac{\text{m}}{\text{s}^2}$  ✓

$$F_g = f_{s\text{max}} \quad f_{s\text{max}} = \mu_s N, \quad N = mg$$

$$f_{s\text{max}} = \mu_s mg$$

$$F_g = \mu_s mg$$

$$\sum \vec{F} = F_g - f_k$$

$$\sum \vec{F} = m\vec{a}$$

$$m\vec{a} = F_g - f_k$$

$$f_k = \mu_k N, \quad N = mg$$

$$f_k = \mu_k mg$$

$$m\vec{a} = \mu_s mg - \mu_k mg$$

$$\vec{a} = \mu_s g - \mu_k g$$

$$\vec{a} = g(\mu_s - \mu_k) = 9.8 \frac{\text{m}}{\text{s}^2} (0.05 - 0.05) = 0 \frac{\text{m}}{\text{s}^2}$$

+3

4. Jessica is producing heat at a rate of about 154 W while doing light exercise. At what rate must water evaporate from her body to get rid of the produced energy? The latent heat of vaporization of water is 2430 kJ/kg at body temperature, 37°C.

$$\dot{Q} = \frac{Q}{t}$$

$$Q = \dot{Q}t$$

$$Q = mL$$

$$\dot{Q}t = mL$$

$$\frac{m}{t} = \frac{\dot{Q}}{L} = \frac{154 \frac{\text{J}}{\text{s}}}{2430 \frac{\text{J}}{\text{g}}} \left| \frac{60 \text{ s}}{\text{min}} \right| = 3.802 \frac{\text{g}}{\text{min}}$$

ANSWER: 3.802  $\frac{\text{g}}{\text{min}}$

$$2430 \frac{\text{kJ}}{\text{kg}} \left| \frac{1000 \text{ J}}{\text{kJ}} \right| \left| \frac{0}{1000 \text{ g}} \right| = 2430 \frac{\text{J}}{\text{g}}$$

5. Paul, who has a mass of 85.0 kg, steps onto a scale that rests on the floor of an elevator. If the elevator accelerates upward at a rate of 7.2 m/s<sup>2</sup>, what will the scale read? Explain why your answer is correct in a couple sentences.

$$\sum \vec{F} = m\vec{a}$$

$$N - W = m\vec{a}$$

$$N = m\vec{a} + mg$$

$$N = m(\vec{a} + g) = 85 \text{ kg} \left( 7.2 \frac{\text{m}}{\text{s}^2} + 9.8 \frac{\text{m}}{\text{s}^2} \right) = 1.445 \times 10^3 \text{ N}$$

ANSWER: 1.445  $\times 10^3 \text{ N}$

a scale measures the normal force. And the normal force needs to be stronger enough than the weight force to produce the desired acceleration on Paul of 7.2  $\frac{\text{m}}{\text{s}^2}$ , so  $N = mg + ma$

Isabella is trying to determine the maximum mass that she can safely hold. Her elbow joint is 40 cm from her bicep, 17 cm from the center of mass of her forearm, and 39 cm from the object. What is the maximum mass that Isabella can hold if, when her bicep is at its limit, her upper-arm bone exerts a force of 1200 N on her elbow joint? Note that the angle between her forearm and her upper arm is 90° and that her forearm has a mass of 2700 g. Note: the forearm is horizontal and the upper arm is vertical.

ANSWER: \_\_\_\_\_

- ★ 7. A freshwater reservoir has a surface area of  $67 \text{ km}^2$  and an average depth of  $60 \text{ m}$ . What mass of water is held behind the dam?

$$\rho = \frac{m}{V} \quad V = Ad$$

$$m = \rho V$$

$$m = \rho Ad = 1000 \frac{\text{kg}}{\text{m}^3} \cdot 67 \text{ km}^2 \cdot \left| \frac{(10^3)^2 \text{ m}^2}{\text{km}^2} \right| \cdot 60 \text{ m} = 4.02 \times 10^{12} \text{ kg}$$

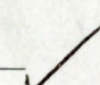
$$\text{ANSWER: } 4.02 \times 10^{12} \text{ kg}$$



8. Emma attaches a nozzle of diameter 5.7 mm to a garden hose of diameter  $2.8 \text{ cm}$ . If the hose fills up a 4-L bucket in  $13.5$  seconds, what is the speed of the water that leaves the nozzle?

$$Q = \frac{V}{t} = \frac{4 \text{ L}}{13.5 \text{ s}} \cdot \left| \frac{\text{m}^3}{1000 \text{ L}} \right| = 2.963 \times 10^{-4} \frac{\text{m}^3}{\text{s}}$$

$$\text{ANSWER: } 11.61 \frac{\text{m}}{\text{s}}$$



$$Q = A v$$

$$v = \frac{Q}{A}$$

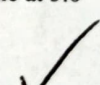
$$v = \frac{Q}{\pi r^2}$$

$$v = \frac{4Q}{\pi d^2} = \frac{4 \cdot 2.963 \times 10^{-4} \frac{\text{m}^3}{\text{s}}}{\pi (0.0057 \text{ m})^2} = 11.61 \frac{\text{m}}{\text{s}}$$

9. Natalie is driving in her car wondering how long would it take her, starting from  $6.1 \text{ m/s}$  and accelerating in a straight line at  $5.0 \text{ m/s}^2$ , to travel a distance of  $710 \text{ m}$ ?

$$\Delta \vec{x} = \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$$

$$\text{ANSWER: } 15.676 \text{ s}$$



$$\frac{1}{2} \vec{a} t^2 + \vec{v}_0 t - \Delta \vec{x} = 0$$

$$t = \frac{-\vec{v}_0 \pm \sqrt{\vec{v}_0^2 - 4 \cdot \frac{1}{2} \vec{a} \cdot -\Delta \vec{x}}}{2 \cdot \frac{1}{2} \vec{a}} \quad \text{quadratic formula}$$

$$t = \frac{-\vec{v}_0 \pm \sqrt{\vec{v}_0^2 + 2 \vec{a} \Delta \vec{x}}}{\vec{a}} = \frac{-6.1 \frac{\text{m}}{\text{s}} \pm \sqrt{(6.1 \frac{\text{m}}{\text{s}})^2 + 2 \cdot 5 \frac{\text{m}}{\text{s}^2} \cdot 710 \text{ m}}}{5 \frac{\text{m}}{\text{s}^2}} = 15.676 \text{ s}$$

time not negative, so + not -

+ 3

10. As Sophia prepares breakfast, she decides to conduct a quick experiment which reveals that the friction force between her steel spatula and her Teflon-coated frying pan is only 1.27 N as she moves the spatula. As she moves the spatula across the surface of the pan, what is the normal force that is exerted on her spatula? Note that for steel on Teflon,  $\mu_s = 0.04$  and  $\mu_k = 0.04$ .

$$f_k = \mu_k N$$

$$N = \frac{f_k}{\mu_k} = \frac{1.27 \text{ N}}{0.04} = 31.75 \text{ N}$$

ANSWER: 31.75 N

11. Isabella throws a 2.9-g coin straight downward from a 24-story window with a vertical speed of 7.1 m/s (approximately 15.9 mph). Ignoring the effect of the drag force, at what speed does the coin impact a patch of grass on the ground? Express the speed in meters per second. Take a story as roughly 10.3 feet. There are 3.28 feet in one meter.

$$24 \text{ story} \left( \frac{10.3 \text{ ft}}{\text{story}} \right) \left( \frac{1 \text{ m}}{3.28 \text{ ft}} \right) = 75.366 \text{ m}$$

ANSWER: 39.084  $\frac{\text{m}}{\text{s}}$

$$v^2 = v_0^2 + 2\vec{a}\Delta\vec{x}$$

$$v = \sqrt{v_0^2 + 2\vec{a}\Delta\vec{x}} = \sqrt{(-7.1 \frac{\text{m}}{\text{s}})^2 + 2 \cdot -9.8 \frac{\text{m}}{\text{s}^2} \cdot -75.366 \text{ m}} = 39.084 \frac{\text{m}}{\text{s}}$$

12. The magnitude of a velocity vector is 13.9 m/s and its x component's magnitude is 4.6 m/s. What is the magnitude of its y component?

$$v^2 = v_x^2 + v_y^2$$

ANSWER: 13.117  $\frac{\text{m}}{\text{s}}$

$$v_y = \sqrt{v^2 - v_x^2} = \sqrt{13.9^2 \frac{\text{m}^2}{\text{s}^2} - 4.6^2 \frac{\text{m}^2}{\text{s}^2}} = 13.117 \frac{\text{m}}{\text{s}}$$

+3

X6

13. Emily drives a 4800-kg truck and descends a 43-m-tall hill at a constant speed. What is the final temperature of 280 kg of brake material if its temperature is 13°C at the top of the hill? Assume the specific heat of the brake material is 830 J/kg°C. Assume the brakes do all the work to maintain the speed.

ANSWER: \_\_\_\_\_

2

14. Ethan drives his 1210-kg car around a 240-m radius unbanked curve at 28.3 m/s (about 63 mph). Ethan's mass is 92 kg. Calculate the centripetal force exerted on the car as Ethan drives around the curve.

$$\vec{F}_c = m \vec{a}_c$$

$$a_c = \frac{v^2}{r}$$

$$1210 \text{ kg} + 92 \text{ kg} = 1302 \text{ kg}$$

ANSWER:  $4.358 \times 10^3 \text{ N}$

$$\vec{F}_c = \frac{mv^2}{r} = \frac{1302 \text{ kg} \cdot 28.3^2 \frac{\text{m}^2}{\text{s}^2}}{240 \text{ m}} = 4.358 \times 10^3 \text{ N}$$

✓

X2

15. Rachel places a ladder of mass 17 kg against her house. The feet of the ladder are resting on her concrete patio at a distance of 7.0 m from the house. The top of the ladder rests against a plastic rain gutter. The friction between the ladder and gutter is negligible. The top of the ladder is at a height of 7.2 m above the concrete patio. The ladder's center of gravity is at its midpoint. Rachel is standing on a rung at a horizontal distance of 4.4 m from the house. If Rachel's mass is 70 kg, what is the minimum static-friction coefficient that would prevent the feet of the ladder from slipping on the concrete?

ANSWER: \_\_\_\_\_

3

41

- ✓ 16. In a particular year, the main span of San Francisco's Golden Gate Bridge was 1277-m long on the coldest day. The bridge was exposed to temperatures ranging from  $-3^{\circ}\text{C}$  to  $41^{\circ}\text{C}$ . What is the change in its length between these temperatures? Assume that the bridge is made entirely of steel so that its coefficient of linear expansion is  $12 \times 10^{-6} \text{ } 1/^{\circ}\text{C}$ .

$$\Delta L = \alpha L \Delta T$$

ANSWER: 0.6743 m

$$L - L_0 = \alpha L_0 \Delta T$$

$$\Delta L = \alpha L \Delta T = 12 \times 10^{-6} \frac{1}{^{\circ}\text{C}} \cdot 1277 \text{ m} \cdot (41^{\circ}\text{C} - (-3^{\circ}\text{C})) = 0.6743 \text{ m}$$

- ✓ 17. Sean's doctor performs a special test and tells him that a 3.3-N force can rupture his eardrum. If his eardrum has an area of  $0.98 \text{ cm}^2$ , at what depth in freshwater would his eardrum rupture, assuming the gauge pressure in the middle ear is zero and he does nothing to alleviate the pressure?

ANSWER: 3.436 m

$$P_g = P_{\text{at}} + \rho g h$$

$$P_g = \rho g h \quad \because P_{\text{at}} = 0$$

$$P = \frac{F}{A}$$

$$\frac{F}{A} = \rho g h$$

$$h = \frac{F}{A \rho g} = \frac{3.3 \text{ N}}{0.98 \text{ cm}^2 \cdot 1000 \frac{\text{kg}}{\text{m}^3} \cdot 9.8 \frac{\text{m}}{\text{s}^2}} \cdot \frac{(10^{-2})^2 \text{ m}^2}{\text{cm}^2} = 3.436 \text{ m}$$

18. Benjamin is playing tennis when he throws a 55-g tennis ball up in the air, hitting it when it's at its peak to give it a speed of  $43.3 \text{ m/s}$ ,  $1.9^{\circ}$  above the horizontal. What is the magnitude of the average force that Benjamin's racket exerts on the tennis ball, given that the ball remained in contact with the racket for  $6.9 \text{ ms}$ ?

ANSWER: \_\_\_\_\_

4

+2

19. Robert's 70-L (18.5-gal) steel gasoline tank is full of gas. Both the tank and the gasoline have an initial temperature of  $9.4^\circ\text{C}$ . After both the tank and the gasoline warm to  $39.9^\circ\text{C}$ , what is the pressure in the gasoline tank if the gasoline cannot spill out? The coefficient of volume expansion for steel is  $35 \times 10^{-6} \text{ } 1^\circ\text{C}$  and for gasoline it is  $950 \times 10^{-6} \text{ } 1^\circ\text{C}$ . Assume the bulk modulus  $B$  for gasoline is  $1.00 \times 10^9 \text{ N/m}^2$ . Show all work.

ANSWER: \_\_\_\_\_

5

20. Brian, a 101-kg ice-hockey goalie who is originally at rest, catches a 0.14-kg ice-hockey puck that is traveling at a velocity of 38.9 m/s. What is the net change in Brian's and the puck's kinetic energy during the collision? Assume friction is negligible.

$$\Delta \vec{p} = 0$$

$$m' \vec{v}' = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

$$\vec{v}' = \frac{m_2 \vec{v}_2}{m'} \quad \because \vec{v}_1 = 0$$

$$m' = m_1 + m_2 = 101 \text{ kg} + 0.14 \text{ kg} = 101.14 \text{ kg}$$

$$\Delta KE = KE_f - KE_i$$

$$\Delta KE = \frac{1}{2} m' (v')^2 - \frac{1}{2} m_1 v_1^2 - \frac{1}{2} m_2 v_2^2$$

$$\Delta KE = \frac{1}{2} m' \left( \frac{m_2 v_2}{m'} \right)^2 - \frac{1}{2} m_2 v_2^2 \quad \because v_1 = 0$$

$$\Delta KE = \frac{1}{2} \cdot 101.14 \text{ kg} \left( \frac{0.14 \text{ kg} \cdot 38.9 \frac{\text{m}}{\text{s}}}{101.14 \text{ kg}} \right)^2 - \frac{1}{2} \cdot 0.14 \text{ kg} (38.9 \frac{\text{m}}{\text{s}})^2 = -105.78 \text{ J}$$

ANSWER: -105.78 J

✓

21. Charles, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.9 cm from the reel's axis of rotation. The reel is given an angular acceleration of  $141 \text{ rad/s}^2$  for 2.2 s. What is the final angular velocity of the reel?

$$\omega = \omega_0 + \alpha t = 0 + 141 \frac{\text{rad}}{\text{s}^2} \cdot 2.2 \text{ s} = 310.2 \frac{\text{rad}}{\text{s}}$$

ANSWER: 310.2  $\frac{\text{rad}}{\text{s}}$

✓

+2

22. If a marathon runner averages 125 furlongs per deciday, then how long would it take him to run a 7.4-league marathon? Report your answer in minutes. A furlong is 10 chains, a mile is 1760 yards, a chain is 22 yards, a day is 10 decadays, and a league is 3 miles.

$$\vec{v} = \frac{\Delta \vec{x}}{\Delta t}$$

ANSWER: 204,595 min

$$\Delta t = \frac{\Delta \vec{x}}{\vec{v}} = \frac{7.4 \text{ league} \cdot \text{deciday}}{125 \text{ furlong}} \cdot \frac{1 \text{ furlong}}{10 \text{ chain}} \cdot \frac{1 \text{ chain}}{22 \text{ yd}} \cdot \frac{1760 \text{ yd}}{1 \text{ mile}} \cdot \frac{3 \text{ mile}}{1 \text{ league}} \cdot \frac{1 \text{ day}}{10 \text{ deciday}} \cdot \frac{24 \text{ hour}}{1 \text{ day}} \cdot \frac{60 \text{ min}}{1 \text{ hour}} = 204,595 \text{ min}$$

23. Brandon is an astronaut walking on the surface of the planet Nova Terra. He experiences a small acceleration due to Earth's gravitational field, even at the distance between the center of Earth and Brandon of 0.26 Gm. What is his acceleration due to the force exerted by Earth's gravitational field alone?

$$G_m = 10^9 \text{ m}$$

$$F = ma$$

$$F_g = G \frac{mM}{r^2}$$

$$ma = G \frac{mM}{r^2}$$

$$a = G \frac{M}{r^2} = 6.673 \times 10^{-11} \frac{\text{N} \cdot \text{m}^2}{\text{kg}^2} \cdot \frac{5.98 \times 10^{24} \text{ kg}}{(0.26 \times 10^9 \text{ m})^2} = 5.903 \times 10^{-3} \frac{\text{m}}{\text{s}^2}$$

ANSWER:  $5.903 \times 10^{-3} \frac{\text{m}}{\text{s}^2}$

24. Adam is spinning a large grindstone by placing his hand on its edge and exerting a force through part of a revolution. How much work does he do if he exerts a force of 95 N through a rotation of 1.0 rad (57°)? He exerts the force perpendicular to the grindstone's 52-cm radius and the effects of friction are negligible. Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is  $(1/2)MR^2$ .

$$W = \tau \theta$$

$$\tau = rF \text{ (perpendicular)}$$

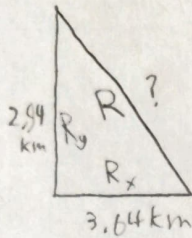
ANSWER: 49.4 J

$$W = rF\theta = 0.52 \text{ m} \cdot 95 \text{ N} \cdot 1.0 = 49.4 \text{ J}$$

+3

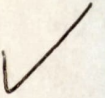
25. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Brian's. To get to Brian's, you have to walk 3.64 km west and then 2.94 km north. How far does he live from you if you drew a straight line on the map?

ANSWER: 4.679 km



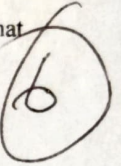
$$R^2 = R_x^2 + R_y^2$$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{3.64^2 \text{ km}^2 + 2.94^2 \text{ km}^2} = 4.679 \text{ km}$$



26. Scott is sitting in his folding chair on the beach, watching his friend Allison as she rides in a speedboat moving at a velocity of 7.81 m/s at an angle of 80.1° north of east. Allison then passes her sister, Ashley, who is riding in a speedboat, which, from Allison's perspective, is travelling at a velocity of 2.19 m/s at an angle of 86.3° west of south. From Scott's perspective, in what direction is Ashley's speedboat travelling?

ANSWER: \_\_\_\_\_



27. Anna attaches a nozzle of diameter 6.7 mm to a garden hose of diameter 2.3 cm. If the hose fills up a 4-L bucket in 11.4 seconds, calculate the pressure in the hose assuming level, frictionless flow.

$$P_2 = P_{\text{atm}} \quad \downarrow \quad \text{no } \rho gh \text{ terms}$$

ANSWER:  $1.5050 \times 10^5 \text{ Pa}$

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$Q = \frac{4 \text{ L}}{11.4 \text{ s}} = \frac{1 \text{ m}^3}{1000 \text{ L}} = 3.509 \times 10^{-4} \frac{\text{m}^3}{\text{s}}$$

$$P_1 = P_2 + \frac{1}{2} \rho (v_2^2 - v_1^2)$$

$$P_1 = P_2 + \frac{1}{2} \rho \left[ \left( \frac{4Q}{\pi d_2^2} \right)^2 - \left( \frac{4Q}{\pi d_1^2} \right)^2 \right] = 101325 \text{ Pa} + \frac{1}{2} \cdot 1000 \frac{\text{kg}}{\text{m}^3} \left[ \left( \frac{4 \cdot 3.509 \times 10^{-4} \frac{\text{m}^3}{\text{s}}}{\pi \cdot 0.0067 \text{ m}} \right)^2 - \left( \frac{4 \cdot 3.509 \times 10^{-4} \frac{\text{m}^3}{\text{s}}}{\pi \cdot 0.023 \text{ m}} \right)^2 \right]$$

$$P_1 = 1.5050 \times 10^5 \text{ Pa}$$

+2

28. Calculate the work output of a Carnot engine operating between temperatures of 681 K and 224 K for 1530 J of heat transfer to the engine.

$$\eta = \frac{W}{Q_h}$$

$$\eta_c = 1 - \frac{T_c}{T_h}$$

$$\frac{W}{Q_h} = 1 - \frac{T_c}{T_h}$$

$$W = Q_h \left( 1 - \frac{T_c}{T_h} \right) = 1530 \text{ J} \left( 1 - \frac{224 \text{ K}}{681 \text{ K}} \right) = 1.0267 \times 10^3 \text{ J}$$

ANSWER:  $1.0267 \times 10^3 \text{ J}$

29. Rachel is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters to a straight ramp. Rachel plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 4.6 m/s. If the height of the quarterpipe is 4.7 m, what must the height of the straight ramp be? Assume that the wheel bearings are well-made and lubricated, so friction can be neglected.

$$\Delta E = 0$$

$$E_1 = E_2$$

$$PE_1 + KE_1 = PE_2 + KE_2$$

$$mgh_1 + \frac{1}{2}mv_1^2 = mgh_2 + \frac{1}{2}mv_2^2$$

$$gh_1 = gh_2 + \frac{1}{2}v_2^2 \quad \because v_1 = 0$$

$$h_2 = \frac{gh_1 - \frac{1}{2}v_2^2}{g} = \frac{9.8 \frac{\text{m}}{\text{s}^2} \cdot 4.7 \text{ m} - \frac{1}{2} \cdot 4.6^2 \frac{\text{m}^2}{\text{s}^2}}{9.8 \frac{\text{m}}{\text{s}^2}} = 3.620 \text{ m}$$

ANSWER: 3.620 m

30. A 57.0-kg block is resting on a so-called "heavy-duty" shelf at a height of 3.5 m above the ground. The shelf suddenly collapses and the block falls to the ground. What is the block's total mechanical energy at a height of 1.8 m, assuming negligible air resistance?

$$\Delta E = 0$$

$$E_1 = E_2$$

$$E_1 = PE_1 \quad \because KE_1 = 0$$

$$E_2 = PE_2$$

$$E_2 = mgh_1 = 57 \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 3.5 \text{ m} = 1.955 \times 10^3 \text{ J}$$

ANSWER:  $1.955 \times 10^3 \text{ J}$

+3

1. 2 J
2. 16.2 MN
3.  $0.0 \text{ m/s}^2$
4. 3.8 g/min or (63, 2) mg/s
5. 1445 N
6. 12.991 kg
7.  $4.02 \times 10^{12} \text{ kg}$
8. 11.61 m/s
9. 15.7 s
10. 31.8 N
11. 39.1 m/s
12. 13.1 m/s
13.  $21.7^\circ\text{C}$
14. 4.34 kN
15. 0.39
16. 67.43 cm
17. 3.44 m
18. 345 N
19. 27.91 MPa
20. -106 J
21. 310.2 rad/s
22. 205 minutes
23.  $5.9 \text{ mm/s}^2$
24. 49.4 J
25. 4.7 km
26.  $6.4^\circ$  west of north
27. 150.49 kPa
28. 1026.74 J
29. 3.62 m
30. 1955 J